Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Data Analyst with 8+ years of production engineering and **hands-on income fraud detection in regulated affordable housing**. At Entrata, reviewed and verified income documentation across Section 8/HCV, LIHTC, and HUD portfolios — detecting income misrepresentation and household-composition discrepancies within fair-housing constraints — then encoded that same decision logic into automated compliance pipelines across 100K+ unit portfolios (zero violations). Applied **Isolation Forest anomaly detection** across 3 research pipelines at George Mason University, surfacing anomalies manual review missed. Achieved $R^2 = 0.977$ regression accuracy across 190+ countries (Python, R). Built CrashClarity — a Random Forest + Logistic Regression pipeline on 104K+ accident records (85%+ accuracy). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

WORK EXPERIENCE

Entrata

Jan 2017 – Aug 2024

Software Engineer — Data Analytics & Fraud Detection

- Reviewed and verified income documentation for **millions of tenant certifications** across Section 8/HCV, LIHTC, and HUD portfolios — identifying inconsistencies between stated income and supporting documents that flagged ineligible or misrepresented applications for escalation.
- Applied **LIHTC/Section 42 and HUD income-limit rules** to determine household eligibility across 100K+ units and multiple jurisdictions — detected income- and household-composition discrepancies (the documented tenant-fraud pattern in LIHTC) while operating within fair-housing constraints requiring uniform, document-driven verification.
- Eliminated **80% of manual eligibility review workload** with zero compliance violations by encoding the decision logic I had applied manually into automated Section 8/HCV, LIHTC, and HUD compliance pipelines — AMI threshold calculations, recertification workflows, and escalation routing across 100K+ unit portfolios.
- Achieved **100% PHI compliance** across all client data by migrating to AWS S3 with Libsodium-encrypted pipelines — zero security incidents; improved SQL query performance **30%** across 200+ queries supporting 1M+ property records at 99.5% uptime.

George Mason University

Aug 2024 – Aug 2025

Research Associate — Statistical Modeling · Anomaly Detection · Data Visualization

- Achieved **$R^2 = 0.977$** regression accuracy across 190+ countries by building multi-source prediction models (Python, R) on WHO and UK Road Safety datasets — cross-validated across environmental, demographic, and behavioral features.
- Surfaced anomalies **manual review missed** across 3 research pipelines by applying Isolation Forest — flagged statistically significant outliers in public health and road safety datasets.
- Improved research comprehension **40%** across two cohorts by designing faculty leadership dashboards in Tableau, ggplot2, and Quarto — translating model outputs into decision-ready visualizations.

PROJECTS

CrashClarity: Large-Scale Accident Severity Prediction (104K Records)

Python · R · Scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages — Live Dashboard

- Achieved **85%+ accuracy** on 104K+ UK accident records by building a Random Forest + Logistic Regression ensemble with feature engineering across weather, road, and time-of-day variables — deployed interactive dashboard on GitHub Pages.

Leasel: Automated Income Eligibility & Compliance Audit Pipeline

Python · LangGraph · LLaMA · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_raq

- Automated the income eligibility and household-composition checks I had applied manually at Entrata: LLM extracts income and household data from 1,240 lease PDFs; a **deterministic rules engine** applies LIHTC/Section 42 and HUD limits to flag violations — compliance logic kept out of the model because these determinations have to be exact and auditable.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules and ~40% injected violation rate — pass@k eval harness (pass@1 81.7%, pass@3 88.3%) with regression gate; every determination is traceable to the triggering rule.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Focus: statistical modeling, ML pipelines, anomaly detection, data visualization

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Analytics & ML: Python, R, SQL, Statistical Modeling, Regression Analysis, Random Forest, XGBoost, Logistic Regression, Isolation Forest (Anomaly Detection), Feature Engineering, Scikit-learn, Pandas, NumPy, Cross-Validation, Hypothesis Testing

Data & Visualization: Tableau, Power BI, ggplot2, Quarto, Jupyter, matplotlib, seaborn, Interactive Dashboards, Data Quality, Data Governance, ETL Pipelines

Data Infrastructure: AWS (S3, EC2, RDS, Glue), PostgreSQL, MySQL, SQLite, Docker, CI/CD (GitHub Actions), RabbitMQ, Git, Linux, PL/SQL

Compliance & Security: PHI Compliance, HIPAA, RBAC, Libsodium Encryption, Audit Pipelines, Quality Gates, Zero-Incident Data Migrations, LIHTC/HCV/HUD Regulatory Workflows

Additional: NLP (spaCy, BERT, Transformers, Embeddings), FastAPI, REST APIs, JavaScript, TypeScript, Agile/Scrum (120+ sprints)